

AMEEN MOHEMMED BASHIR

EMBEDDED FIRMWARE ENGINEER

CONTACT

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Kaliyanil (H), Erathuvadakara (P.O),
Vellavoor, Kottayam, Kerala.

SKILLS

- C, C++, Embedded C Development.
- 16, 32 bit Micro controller.
- Communication protocols.
- Debugging using Breakpoints.
- Testing on Dyno and Vehicle.
- Electric Vehicle Power Train.
- Motor control.

TOOLS USED

- Dave, CCS, Picoscope, Ozone.
- µC/Probe, TS master, Bus master,
- PCAN, Keil, Git, Odo

EDUCATION

**B Tech In Electronics and
Communication**

**Rajiv Gandhi Institute Of
Technology, Kottayam**

2019-2022

Course Completed.

**Diploma In Electronics and
Communication**

**Model Polytechnic College,
Mattakara, Kottayam**

2016-2019

Completed Professional Diploma in
Electronics and Communication
Engineering with 8.1 GPA.

LANGUAGES

English

Tamil

Malayalam

PROFILE

Embedded Firmware Engineer with experience in firmware development life cycle: specification, design, implementation, integration, testing, troubleshooting, and documentation of embedded systems. Expertise in applying system architecture and HW/SW interaction knowledge to maximize the overall performance under resource limitations of real-time embedded systems.

WORK EXPERIENCE

Embedded Engineer

Nhanz Systems Pvt Ltd.

2023 May - Present

- Hands-on experience in developing and testing BSW and ASW for TI and Infineon micro controllers.
- Tested and Tuned Field-Oriented Control (FOC) algorithms for diverse motor types including induction motors and Interior Permanent Magnet Synchronous Motors (IPMSM).
- Tested CAN communication, and modified CAN DBC files to meet customer requirements.
- Conducted dyno testing on various types of motors including IPMSM, induction motors, and hub motors .
- Prepared technical reports and communicated with hardware design team.
- Collaborated with the hardware team and prepared testing codes to test the hardware.
- Worked collaboratively in a team environment and shared knowledge and expertise
- Adapted to new technologies and methodologies in my work.

Junior Embedded Engineer

Nhanz Systems Pvt Ltd.

2022 July - 2023 May

- Hands-on experience in developing and debugging various types of micro-controller peripherals, including ADC, PWM, communication peripherals , POSIF, capture modules, EQEP, and more.
- Interfaced various types of position sensors including sin/cos encoders, UVW hall encoders, ABZ encoders, etc.

PROJECTS

Two, three and four wheeler vehicle function implementation

- Implemented Torque Co-ordination block for 2, 3 and 4 wheeler vehicle and tested it on Dyno and Jack up Vehicle.
- Tested CAN communication and modified the DBC based on customer requirements.
- Implemented and tested Pheripherals such as POSIF, Timers .
- Conducted dyno testing, including four-quadrant testing for forward drive, reverse drive, forward regeneration, and reverse regeneration.

HUB Motor Controller

- Developed and tested a library for interfacing with UVW hall encoders to measure rotor position.

Induction Motor Controller

- Developed and tested a library for interfacing with ABZ encoders to measure rotor speed.

Interior Permanent Magnet Synchronous motor controller

- Tested and tuned Field Oriented Control (FOC) algorithm .
- Conducted dyno testing, including four-quadrant testing such as forward drive, reverse drive, forward regeneration, and reverse regeneration.
- Tested and tuned speed control for an IPMSM (Interior Permanent Magnet Synchronous Motor).